

# Extending InternetSim for Internet-Scale BGP Security Analysis

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Route Origin Validation | BGP Hijacking | Autonomous Systems

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*Technical Research Report*

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76,802 ASes | 495,579 Links | Internet-Scale Topology

## Abstract

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*The Border Gateway Protocol (BGP) is the backbone of inter-domain routing on the Internet. However, BGP is vulnerable to security threats such as prefix hijacking due to the lack of built-in authentication. Route Origin Validation (ROV) has been proposed as a mitigation mechanism, but its effectiveness depends on deployment strategies. In this project, we extend InternetSim, an Internet-scale inter-domain routing simulator, by adding performance metrics and an interactive command-line interface. Using a real-world topology with over 76,000 Autonomous Systems (ASes) and 495,000 links, we conduct experiments to analyze BGP convergence, hijacking impact, and the effectiveness of different ROV deployment strategies. Our results show that targeted deployment of ROV on high-degree ASes significantly reduces hijacking impact compared to random deployment.*

## 1. Introduction

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The Internet is composed of thousands of interconnected networks known as Autonomous Systems (ASes). These ASes exchange routing information using the Border Gateway Protocol (BGP), which determines how data packets traverse the global Internet.

Despite its critical role, BGP lacks intrinsic security mechanisms, making it vulnerable to attacks such as prefix hijacking. In a hijack, a malicious AS falsely announces ownership of IP prefixes, causing traffic to be misrouted to unintended or malicious destinations.

To address this vulnerability, Route Origin Validation (ROV) has been proposed. ROV enables ASes to verify whether a route announcement is legitimate using cryptographic information stored in the Resource Public Key Infrastructure (RPKI). However, ROV deployment across the Internet is still incomplete, and its effectiveness depends critically on how widely and strategically it is adopted.

In this project, we extend InternetSim to analyze BGP behavior and evaluate the impact of ROV deployment strategies at Internet scale. We introduce new performance metrics, implement an interactive command-line interface (CLI), and conduct large-scale experiments to derive meaningful insights about routing security.

## 2. Background

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### 2.1 Border Gateway Protocol (BGP)

BGP is a path-vector routing protocol used to exchange routing information between ASes. Each route advertisement contains key attributes including:

- AS Path — the sequence of ASes a route has traversed
- Local Preference — a metric used by ASes to prefer certain routes over others
- MED (Multi-Exit Discriminator) — used to influence inbound traffic from neighboring ASes
- Next-Hop — the next router to forward traffic toward the destination prefix

BGP routers select the best route among competing advertisements based on these attributes and forward that route to neighbors. BGP was originally designed for reachability and scalability, not security.

### 2.2 BGP Hijacking

BGP hijacking occurs when an AS announces a prefix it does not legitimately own. Since BGP does not natively verify the legitimacy of route announcements, other ASes may accept and propagate the false route. This can result in:

- Traffic interception — where packets are redirected through the attacker's network
- Traffic blackholing — where packets are dropped entirely
- Man-in-the-middle attacks — where traffic is inspected before being forwarded

Notable real-world incidents include the 2010 China Telecom event, where approximately 15% of Internet routes were briefly hijacked, and the 2018 BGP hijack affecting Amazon Route 53 DNS infrastructure.

### 2.3 Route Origin Validation (ROV)

ROV is a security mechanism that allows ASes to validate route announcements using the Resource Public Key Infrastructure (RPKI). Each legitimate route origin is covered by a Route Origin Authorization (ROA) — a cryptographically signed object asserting which AS is authorized to originate a given IP prefix.

When an AS receives a BGP announcement, ROV allows it to check whether the announcement matches a valid ROA. Announcements that are invalid (wrong origin AS or wrong prefix length) can be filtered out, reducing the propagation of hijacked routes.

## 3. System Overview

InternetSim is an Internet-scale inter-domain routing simulator designed to efficiently simulate BGP behavior across large topologies. It supports realistic AS relationship datasets and allows users to define routing scenarios through two distinct operating modes: file-based batch execution and an interactive command-line interface.

### 3.1 Operating Modes

The simulator supports two modes of operation, selectable at startup:

- **Mode 1 — Description File (descriptionFile.txt):** In this mode, the simulator reads all routing commands from a pre-written description file (typically descriptionFile.txt). The path to this file is configurable via the config file, allowing different experiment scripts to be run without modifying source code. This mode is suited for reproducible, automated, large-scale experiments where the full sequence of events is known in advance.
- **Mode 2 — Interactive CLI:** In this mode, the simulator launches an interactive terminal session. The user issues commands in real time, allowing dynamic experimentation, ad-hoc queries, and exploratory analysis. The CLI mode supports all the same operations as the description file, but executed interactively with immediate feedback after each command.

### 3.2 Data Loading and Configuration

Regardless of the selected mode, the simulator must load topology and configuration data before processing routing events. Data loading behavior depends on what is specified in the config file:

- **Config file pre-configured:** If the config file specifies the AS relationship dataset and relevant file paths, the simulator (in CLI mode) can load all data automatically using the init command. This initializes the full topology without requiring the user to specify file paths manually.
- **CLI override:** In CLI mode, the user can explicitly load or overwrite any data source directly from the terminal — including the AS relationship file, ROV configuration, and other parameters — regardless of what the config file specifies. This allows flexible experimentation without editing configuration files between runs.

This two-level configuration model (config file defaults + CLI overrides) provides a balance between convenience for standard experiments and flexibility for exploratory or custom scenarios.

### 3.3 Input

- AS relationship dataset — comprising 76,802 ASes and 495,579 inter-AS links
- descriptionFile.txt — batch command script for file-based mode
- Config file — specifies file paths, mode selection, and default parameters
- CLI commands — including init, announce, withdraw, and policy changes
- ROV configuration — specifying which ASes validate route origins

### 3.4 Processing

- BGP route propagation — implementing valley-free routing policy
- Policy evaluation — customer, peer, and provider relationship handling
- Event-driven simulation engine — processing routing events in order
- ROV filtering — dropping invalid route announcements at ROV-enabled ASes

### 3.5 Output

- Routing tables — for each AS in the simulated network
- Simulation metrics — convergence time, update count, affected AS count
- Interactive CLI output — real-time feedback during experimentation

## 4. Contributions

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The following extensions were implemented as part of this project:

### 4.1 Performance Metrics

Three key metrics were added to quantify simulation behavior:

- **Convergence Time:** measures the wall-clock time from the first routing event to the point at which no further BGP updates are generated. This reflects how quickly the network stabilizes.
- **Total BGP Update Count:** tracks the total number of routing messages exchanged during the simulation. This serves as a proxy for protocol overhead and computational load.
- **Number of Affected ASes:** counts the number of ASes whose routing tables change during the simulation, indicating the scope of propagation.

### 4.2 Simulator Input Modes

InternetSim supports two operational modes for loading and executing simulation scenarios:

#### Mode 1: Description File

The simulator can be configured to load a description file (e.g., description.txt) automatically at startup via the config file. All routing commands, announcements, and policy directives are read from this file and executed in sequence without any user interaction.

## Mode 2: Interactive CLI

The CLI mode allows real-time, interactive experimentation. It supports the following workflow:

- **init command:** loads and executes the description file(s) specified in the config file, initializing the simulation state.
- **File overriding:** before running init, users can load or overwrite the description file directly from the CLI, giving fine-grained control over which scenario is initialized.
- **reload command:** loads additional description files on top of the current simulation state without resetting existing announcements, allowing incremental scenario building.
- **Reset behavior:** if the simulator is restarted or init is re-run without reload, all previous announcements are cleared and the state resets from scratch.

This dual-mode design allows InternetSim to be used both as a batch simulation tool (via description files) and as an interactive research platform (via CLI), with flexible file management in both modes. A detailed User Guide covers the full command reference.

Detailed CLI documentation and usage examples are available in the project repository.

 [CLI\\_Documentation](#) [Demo](#)

## 4.3 Experimental Framework

A structured experimental framework was developed to support reproducible large-scale experiments, including:

- BGP hijack simulation — modeling attacker-originated prefix announcements
- ROV deployment analysis — comparing random vs. targeted deployment strategies
- Large-scale evaluation — running experiments across the full 76K-AS topology

## 5. Experimental Setup

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All experiments were conducted using the following configuration:

- Dataset: CAIDA AS-relationship data comprising 76,802 ASes and 495,579 links
- Environment: Python-based InternetSim simulation engine
- ROV Strategies: None, random (5%, 10%, 20%, 50%), Tier-1 only, high-degree (5%, 10%, 20%)
- Hijack Scenario: Two ASes simultaneously announcing the same IP prefix

## 6. Experiments and Results

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### 6.1 Baseline Routing

Initial experiments involved normal prefix announcements from multiple legitimately authorized ASes. The simulator propagated routes across the full network topology and reached stable convergence. Key baseline measurements:

- Convergence time: approximately 5.8 seconds for the full 76K-AS topology
- BGP updates: approximately 1.36 million messages exchanged during propagation
- Affected ASes: all 76,000+ ASes received and processed routing updates

These measurements represent a representative full-scale announcement scenario and are used as a reference for comparison across experiments.

## 6.2 BGP Hijacking Simulation

A hijacking scenario was simulated by introducing a malicious AS that announces a prefix it does not own, in competition with the legitimate origin AS. Key observations from this experiment:

- A significant portion of ASes selected the malicious route, depending on their topological proximity to the attacker
- The hijack generated a high number of BGP updates as competing routes propagated through the network
- ASes that were topologically closer to the attacker were more likely to accept the hijacked route
- The experiment confirmed BGP's fundamental vulnerability: no verification is performed on the origin of route announcements

This validates the need for origin validation mechanisms such as ROV to mitigate the impact of prefix hijacking.

## 6.3 ROV Deployment Analysis

The core experiment evaluated nine different ROV deployment strategies across the same hijacking scenario. Results are summarized in the table below:

| Deployment Strategy | ROV ASes | BGP Updates | Affected ASes | Convergence (s) | Notes            |
|---------------------|----------|-------------|---------------|-----------------|------------------|
| Random 0%           | 0        | 529,289     | 76,382        | 0.8708          | No ROV           |
| Random 5%           | 3,840    | 323,932     | 71,653        | 0.7828          | Random selection |
| Random 10%          | 7,680    | 302,074     | 70,029        | 0.7443          | Random selection |
| Random 20%          | 15,360   | 193,570     | 65,681        | 0.4156          | Random selection |
| Random 50%          | 38,401   | 55,762      | 35,986        | 0.1432          | High deployment  |
| Tier-1              | 37       | 307,409     | 66,053        | 0.7525          | Tier-1 AS only   |
| High-degree 5%      | 3,840    | 831         | 831           | 0.00224         | Top-degree ASes  |
| High-degree 10%     | 7,680    | 823         | 823           | 0.00275         | Top-degree ASes  |
| High-degree 20%     | 15,360   | 818         | 824           | 0.00218         | Top-degree ASes  |

Key observations from the ROV deployment experiments:

- **No ROV (0%):** 529,289 updates and 76,382 affected ASes demonstrate the full impact of an unmitigated hijack.
- **Random deployment:** effectiveness increases gradually with adoption. Reaching 50% random deployment is required to substantially reduce impact (55,762 updates, 35,986 affected ASes).
- **Tier-1 only:** deploying ROV exclusively at the 37 Tier-1 ASes provides only moderate benefit (307,409 updates), similar to random 5% deployment.
- **High-degree targeted:** even 5% targeted deployment at the highest-degree ASes reduces updates to just 831 and affected ASes to 831 — a reduction of over 99.8% compared to no ROV.

Figure 1 describes the behaviour visually for better analysis.

**Fig 1.1** — Affected ASes decrease steadily with random ROV deployment, but even at 50% adoption nearly half the network remains vulnerable, showing the inefficiency of uncoordinated deployment. Less no of AS affected implies better security.

**Fig 1.2** — BGP updates drop sharply at low deployment levels then taper off, confirming that random ROV alone cannot adequately contain a hijack at Internet scale.

**Fig 1.3** — Convergence time decreases consistently with ROV adoption, as early route rejection reduces overall update propagation across the network.

**Fig 1.4** — On a log scale, High-Degree 10% deployment reduces affected ASes, BGP updates, and convergence time by several orders of magnitude compared to Tier-1-only deployment, demonstrating the overwhelming advantage of topology-aware targeting.

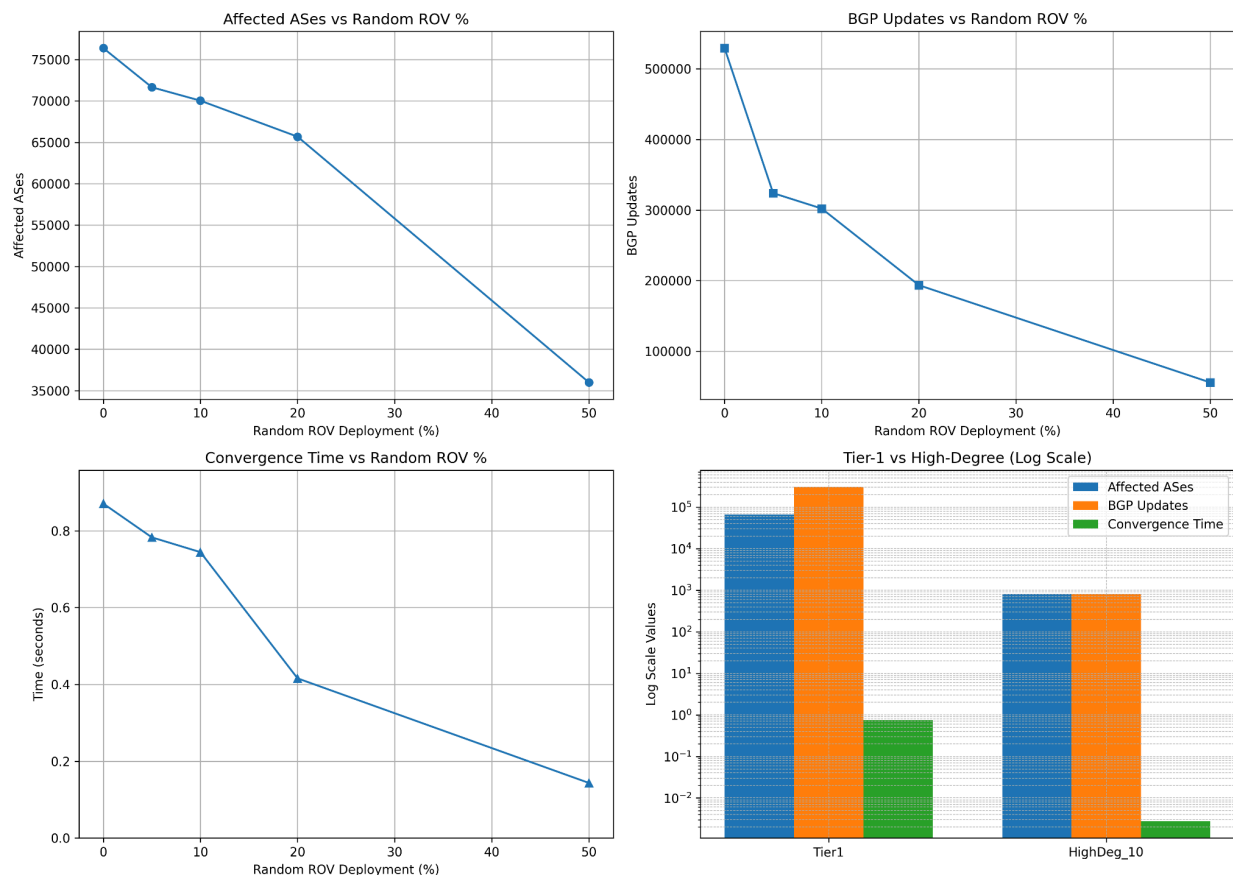


Figure 1 : How random ROV implementation affect the internet

## 6.4 High-Degree ROV Threshold Analysis

Building on the finding that high-degree AS targeting is highly effective, a dedicated threshold experiment was conducted. ROV was progressively enabled on the top-N highest-degree ASes ( $N = 0, 50, 100, \dots 450$ ) to identify the minimum deployment needed to secure 50% and 90% of the network.

The table below shows BGP updates and the percentage of the 76,802-AS network that achieves a secure routing path (i.e., ASes not affected by the hijack):

| Top-N High-Deg ASes | BGP Updates   | Affected ASes | % Network Secured | Milestone     |
|---------------------|---------------|---------------|-------------------|---------------|
| 0                   | 341,452       | 72,435        | 5.7%              |               |
| 50                  | 212,932       | 53,582        | 30.2%             |               |
| 100                 | 173,321       | 49,016        | 36.2%             |               |
| <b>150</b>          | <b>87,441</b> | <b>37,907</b> | <b>50.6%</b>      | <b>≥50% ✓</b> |
| 200                 | 65,199        | 31,108        | 59.5%             |               |
| 250                 | 47,842        | 26,666        | 65.3%             |               |

| Top-N High-Deg ASes | BGP Updates | Affected ASes | % Network Secured | Milestone |
|---------------------|-------------|---------------|-------------------|-----------|
| 300                 | 40,981      | 24,850        | 67.6%             |           |
| 350                 | 36,616      | 22,085        | 71.2%             |           |
| 400                 | 9,874       | 8,344         | 89.1%             |           |
| 450                 | 8,339       | 7,284         | 90.5%             | ≥90% ✓    |

Key thresholds identified:

- **50% network secured:** achieved at ~150 high-degree ASes with ROV enabled (50.6% secured), requiring ROV deployment on just 0.2% of all ASes.
- **90% network secured:** achieved at ~450 high-degree ASes with ROV enabled (90.5% secured), still under 0.6% of all ASes.
- **Non-linear jump at 400 ASes:** a sharp transition occurs between 350 and 400 nodes — network coverage jumps from 71.2% to 89.1%, suggesting a critical connectivity threshold in the AS topology.

These results demonstrate that ROV deployment on fewer than 500 strategically chosen ASes is sufficient to secure over 90% of the global Internet routing infrastructure — a powerful argument for targeted, incentive-driven ROV adoption. Refer Figure 2.

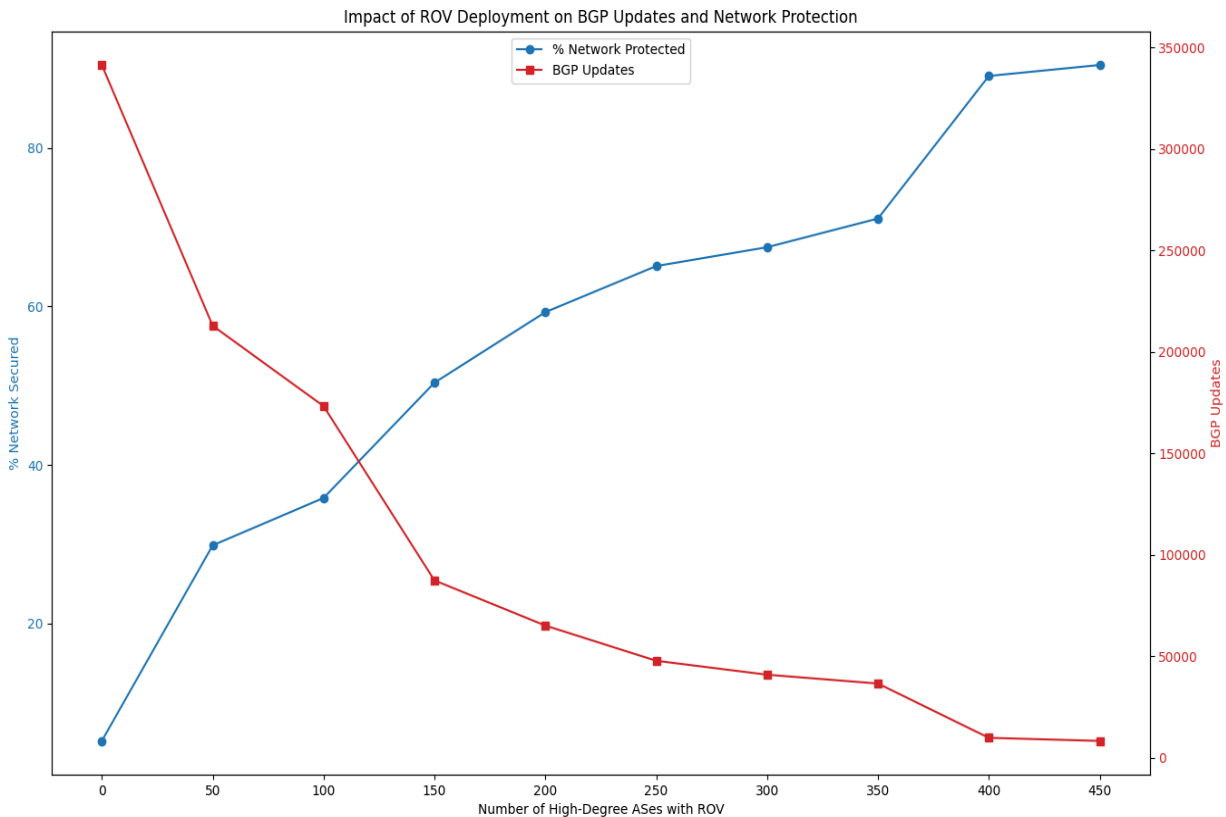


Figure 2

**Fig. 2** — Network protection increases from **5.7% to 90.5%**, while BGP updates drop sharply as more high-degree ASes adopt ROV. A notable tipping point occurs beyond **400 ASes**, where coverage jumps from **71% to 89%**. Out of **76K ASes**, deploying ROV on the top **400 high-degree ASes** reduces hijack impact by **~90%**.

## 7. Key Insights

*Targeted deployment of ROV at high-degree ASes is dramatically more effective than random deployment. Just 5% targeted adoption reduces hijacking impact by over 99%, while 50% random adoption achieves only a 93% reduction.*

- **Deployment scale matters:** increasing the fraction of ROV-enabled ASes consistently reduces both the number of BGP updates and the number of affected ASes, regardless of strategy.
- **Random deployment has diminishing returns:** high random adoption rates are required to approach the effectiveness of small targeted deployments.
- **Topology-aware targeting is critical:** high-degree ASes occupy central positions in the AS graph and therefore influence the greatest number of downstream routing decisions.
- **Tier-1 alone is insufficient:** despite their privileged network positions, the 37 Tier-1 ASes alone do not cover enough of the routing landscape to effectively contain hijacks.
- **Network effect:** ROV effectiveness is not linear — small increases in targeted deployment can yield disproportionately large reductions in hijack impact due to the hierarchical structure of the Internet.

## 8. Challenges Faced

### 1. Python–C++ Integration via Cython

Integrating Python with C++ using Cython was challenging due to differences in memory management and data representation. Handling type conversions and debugging across compiled and interpreted layers required careful coordination.

### 2. Handling Internet-Scale Topology

Simulating a network with 76K+ ASes and ~500K links introduced scalability challenges in terms of memory usage and execution time, where even minor inefficiencies significantly impacted performance.

### 3. Efficient Metric Collection

Incorporating metrics such as BGP updates, affected ASes, and convergence time required modifying the simulation flow while ensuring minimal overhead and avoiding incorrect counting during large-scale propagation.

## 4. Designing Fair and Reproducible Experiments

Ensuring consistency across different ROV deployment strategies (random, Tier-1, high-degree) required careful control of initial conditions to enable accurate and meaningful comparisons.

## 5. Debugging and Interpreting Large-Scale Behavior

Debugging routing behavior across thousands of ASes was complex, as errors propagated widely. Interpreting non-intuitive results required a deeper understanding of network topology and routing dynamics.

## 9. Conclusion

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This project successfully extended InternetSim to support interactive experimentation and fine-grained performance analysis. Through large-scale simulations on a real-world Internet topology of 76,802 ASes and 495,579 links, we demonstrated the practical impact of BGP hijacking and systematically evaluated the effectiveness of diverse ROV deployment strategies.

Our results provide strong evidence that strategic, topology-aware deployment of ROV — particularly targeting high-degree ASes — can significantly improve Internet routing security with minimal adoption overhead. This finding has direct policy implications: rather than requiring universal adoption (which faces significant coordination challenges), focusing deployment effort on the most connected ASes can yield outsized security improvements.

The interactive CLI and performance metric extensions developed in this project provide a valuable foundation for ongoing BGP security research and experimentation at Internet scale.

Github Repository of the Implementation: <https://github.com/codesayan2004/InternetSim>

Coding Documentation: [InternetSim\\_Coding\\_Change\\_Documentation](#)

## 10. Limitations of InternetSim

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1. **Limited Real-World Accuracy:** Cannot capture hidden policies and private AS agreements.
2. **Lock Contention:** Multi-threading may face scalability issues due to shared locks.
3. **No IGP Modeling:** Focuses only on BGP, ignores intra-domain routing.
4. **Simplified Router Model:** Uses a single RIB, missing real router complexities.
5. **Database Dependency:** Performance depends on external storage (e.g., MongoDB).
6. **Limited Scope:** No packet-level simulation or network dynamics (latency, failures).
7. **Static Topology:** Does not model dynamic Internet changes.
8. **High Complexity:** Requires detailed configurations and has a steep learning curve.

## 11. Future Work

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Several promising directions for future research have been identified:

- **Real-time simulation integration:** connecting InternetSim to live BGP feed data to enable real-time monitoring and analysis of routing events on the actual Internet.
- **Visualization tools:** developing graphical tools for visualizing route propagation, hijack spread, and ROV coverage across the AS topology.
- **Advanced attack scenarios:** modeling more sophisticated attacks such as sub-prefix hijacking, path manipulation, and BGP policy manipulation.
- **Policy optimization:** applying graph-theoretic and optimization techniques to identify the minimal set of ASes that should deploy ROV to achieve a target security threshold.
- **Economic modeling:** incorporating cost-benefit analysis to evaluate ROV deployment incentives for individual network operators.

## References

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